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An RFID-based System for Library Management and Its Performance Evaluation

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***Abstract*—Using electromagnetic coupling, an RFID tag can get power supplier by a reader and communicate with it for data exchange. Because the RFID system enables non-contact communication, various services and applications including the management of a library catalogue are possible. However, the system is affected easily by neighboring environment and the resonant frequency, thus the communication performance is degraded. In this paper, is used 13.56 MHz RFID system for the management of the library. We evaluate the influence that papers or other RFID tags give to the resonant frequency of an RFID tag.**

***Keywords*-Radio Frequency Identification; RFID; Automatic Identification; Resonant Frequency**

I. Introduction

The progress of the radio technology allow us to use many wireless services, for example, TV, Radio, mobile phone, etc. Recently, using a radio wave technique to get the information from goods without direct contact is very important. This technique is called Radio Frequency Identification (RFID) [1].

The RFID technique uses electromagnetic coupling for data exchange between the reader/writer and the tag. The RFID system using this technique can be applied in the case that a large quantity of goods is managed. For example, in the library, by using RFID system, we expect the efficiency of the following services: 1) rental of book and the return, 2) collection inventory, 3) search of the book, 4) access control of users [2], [3]. Furthermore, if RFID system is integrated with smart phones and sensor networks [4], [5], as new services for the library system to trace books and users, the system may send useful information to users.

A lot of applications using RFID are proposed [6]–[9]. Moreover, the development of RFID devices and performance evaluation using RFID system are performned to realize reliable RFID systems [10]–[17].

At Kyushu University Library, we have implemented a RFID system. For the evaluation of usefulness of RFID system, we carried out experiments at Chikushi Library of Kyushu University, as a joint study with Mitsubishi Materials Corp. and Checkpoint Systems Inc. [2]. In these experiments, we found that the performance depends on the reading range of RFID system. Especially, when tags get too close, the performance largely decreased. The performance of the tag system changed by the distance of bookshelf or desk. The main reason of this problem is the shift of the resonant frequency by the influence of other things.

In this paper, to confirm this situation, using 13.56 MHz RFID system, we evaluated the influence that the paper or another RFID tag give to the resonant frequency of the RFID tag. The paper structure is as follows. In the next Section, we present the RFID system. Then, we give the evaluation of 13.56 MHz RFID tag. Finally, we conclude the paper.

II. RFID System

An RFID system is contactless ID system using wireless communication. It is one of the technique used for the automatic identification. The automatic identification means to gautomatically input bar-code, magnetic-card, RFID data, etc. with the use of hardware and software and not human intervention in order to recognize the content of the datah. Also, the biometrics, OCR, the machine vision are included in this technique.

An RFID system is made up of two components as shown in Fig. 1 [1]. One is the RFID tag, which is located on the object to be identified, and another is the reader/writer. The RFID tag normally does not have the power supply to work, so the reader/writer not only exchange the data, but also supply the power and clock signal to the RFID tag. To do this, the RFID tag has an antenna or a coil as a coupling element for the communication, as shown in Fig. 2. Figure 3 shows a practical example of label type RFID tag using 13.56 MHz. The small black box inside the coil in this figure is an IC chip. The RFID tag has various shapes, for example, label, card, coin, and stick.

RFID systems are classified according to the operating frequency, the physical coupling method and the communication range. For example, available operating frequency is 135 kHz, 13.56 MHz, 900 MHz, and 2.45 GHz. The operating frequency is used properly depending on a purpose.

III. Evaluation of 13.56 MHz RFID Tag

13.56 MHz RFID systems are widely known and used it for the management of a library catalogue. In Kyushu University, using 13.56 MHz RFID system, the effects of the introduction of the RFID system to the library were evaluated [2]. So, in this paper, 13.56 MHz RFID tag is

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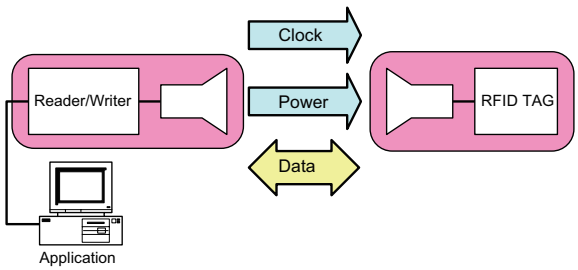

Figure 1. Basic concept of RFID system

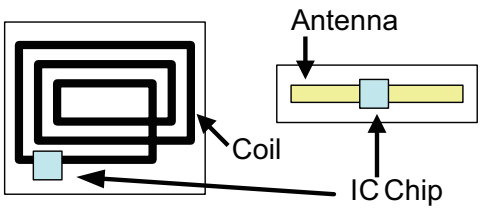

Figure 2. Basic rayout of RFID Tag

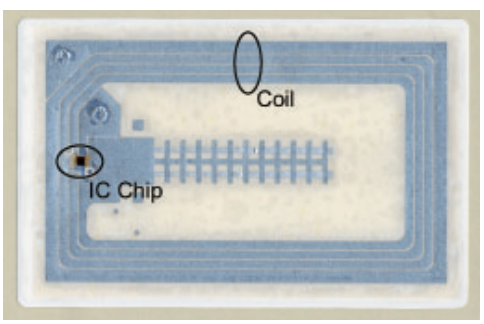

Figure 3. Example of RFID tag in practice use

Table I
TAG SAMPLES USED IN THIS EVALUATION

	Sample A	Sample B	Sample C
Standard	Original	ISO15693	ISO15693
No. of samples	32	10	4
Shape	label	label	label

evaluated. The 13.56 MHz RFID system is based on the international standard of ISO15693. In this paper is used the label and card shape RFID tags shown in Table I and evaluated the resonant frequency of the tag.

A. Resonant Frequency of RFID Tag

13.56 MHz RFID system is connected by the inductive coupling. In this case a tag is combined with a reader in a resonant frequency that corresponds with the transmission frequency of the reader. Generally, RFID tag has the characteristic between induced voltage at a tag coil and frequency as shown in Fig. 4 [18], [19]. In this figure, f_0 is the resonant frequency of tag, f_c is the transmission frequency of the

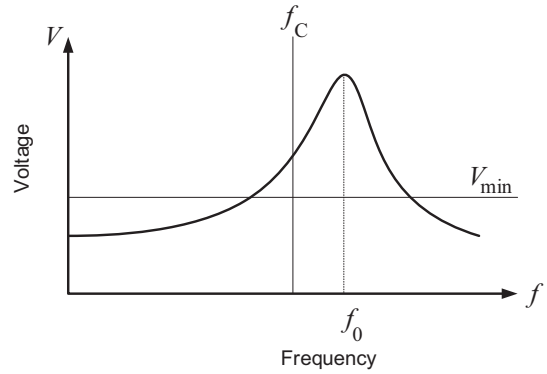

Figure 4. Induced voltage at a tag coil vs frequency

Table II
RESONANT FREQUENCY OF RFID TAG

	Sample A	Sample B	Sample C
Average	14.040	14.333	13.521
Maximum	14.212	14.460	13.546
Minimum	13.992	13.380	13.483
Variance	0.0025	0.0399	0.1542

[MHz]

reader, and $V_{\min}$ is the minimum of the voltage necessary to drive a tag. If the voltage to occur in f_c is lower than $V_{\min}$, the tag does not work. On the other hand, in the case f_0 becomes f_c, the voltage becomes maximum and maximum power is supplied to the tag. In this case because the voltage of f_c is larger than $V_{\min}$, the tag works. To know the characteristics of the resonant frequency of tag in real situation is very important to develop the RFID systems. Using the grid dip oscillator DELICA DMS-230S2, the resonant frequency f_0 of the RFID tag show in Table I was observed. In this experiment is prepared a testing bench as shown in Fig. 5. The observation results are shown in Table II. The measurements for all samples are carried out four times. From the results, the resonant frequency becomes approximately 14 MHz except for sample C. In this situation, because the resonant frequency of sample C is near the frequency of reader, it seems that RFID tag of sample C has the highest performance. However, when the tag is attached on goods such as books, inductive coupling occurs between a tag and goods and the resonant frequency becomes low. To check this effect the board made by plastic which is between a tag and the grid dip oscillator is removed and the resonant frequency is measured. As shown in Table III, the resonant frequency of all tags becomes higher than the results in Table II. Also, the effect of goods to the resonant frequency of tag are clearly shown.

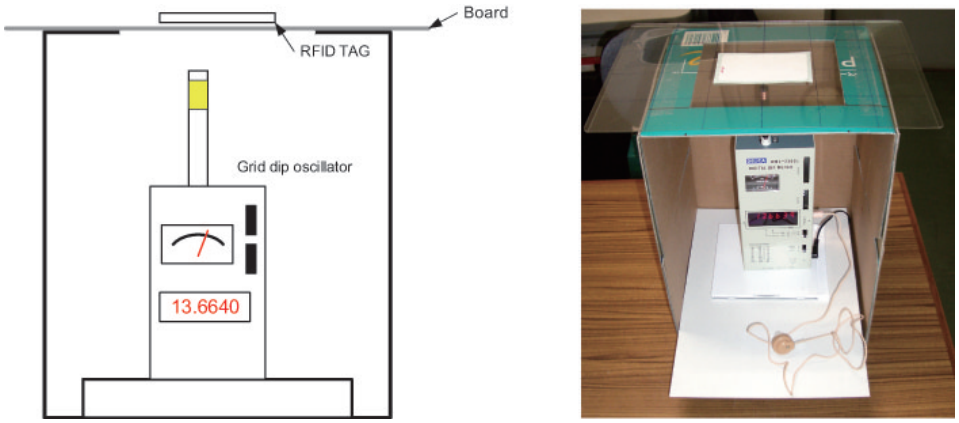

Figure 5. Measurement image of the resonant frequency of RFID tag

Table III
RESONANT FREQUENCY OF RFID TAG IN THE CASE OF THE BOARD BETWEEN TAG AND OCILLATOR REMOVED

	Sample A	Sample B	Sample C
Average	14.112	14.429	13.629
Maximum	14.262	14.498	13.651
Minimum	14.042	14.185	13.601
Variance	0.0018	0.0082	0.0004

[MHz]

Table IV
RESONANT FREQUENCY OF RFID TAG ATTACHED TO BOOK

	Book	Book	Book	Magazine
Size	B5	A4	B5	A4
No. of pages	300	650	175	80
Resonant Frequency	13.682	13.648	13.717	13.902

[MHz]

B. Effect of Goods on Resonant Frequency of RFID Tag

The RFID tag is attached on goods such as books. In this case, the resonant frequency of RFID tag is changed by goods. Table IV shows the resonant frequency of tag attached to books. In this evaluation is used one of the tag in sample A. As shown in this table, the resonant frequency of the tag attached to a book is not the same compared with the case of tag only. As the volume of book increases, the resonant frequency seems to be low. Normally, the goods act as a parasitic capacitance for the RFID tag and the change of the resonant frequency of RFID tag is low. As a trial, a tag was attached on a case of music CD and is measured the resonant frequency of tag. However, we couldnft get the data. It is thought that aluminium depositing onto a CD has a great influence.

Next, by using a testing bench shown in Fig 6 is evaluated the relation between the volume of papers and the resolution frequency of a tag attached on it. In this evaluation, the volume of copying papers of A4 size put on the tag was changed and the resonant frequency of the tag was measured. The tag was selected from sample A. The results are shown in Fig. 7. In this figure, the solid line is the best fit curve obtained by assuming a logarithmic function. Figure 8 shows the best fit curve after the same evaluation using another tag. From these figures, it is shown clearly that, as the volume of copying papers is increased, the resonant frequency is changed like a logarithmic function and finally this resonant frequency converges in a certain value.

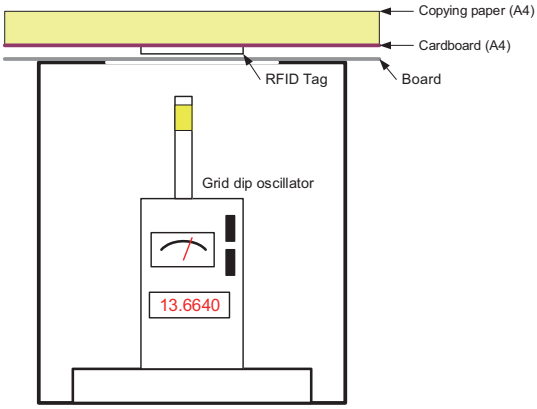

Figure 6. Measurement image of the resonant frequency of RFID tag affected by copying papers

C. Effect of Other Neighboring RFID Tags on Resonant Frequency of Target RFID Tag

If other RFID Tags are near the target RFID tag, the resonant frequency of target tag is changed. For example, the situation that two pieces of tags are adjacent occurs when thin books are stood on the book shelf. This influence is large compared with the influence of other goods such as a book. In order to show this difference clearly, using the method shown in Fig. 9, the relation between the resonant frequency and the distance between two pieces of tags is

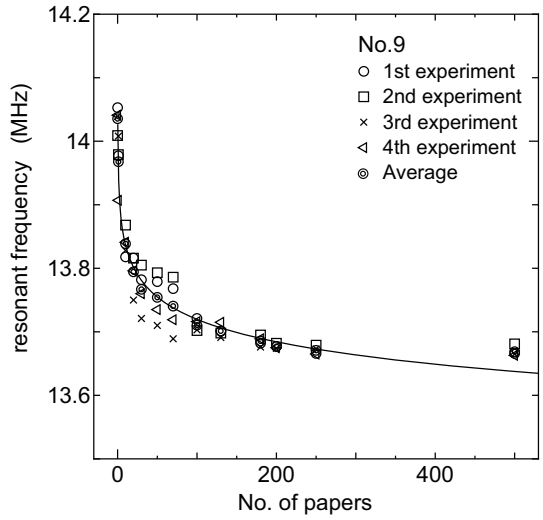

Figure 7. Resonant frequency of RFID tag vs number of papers

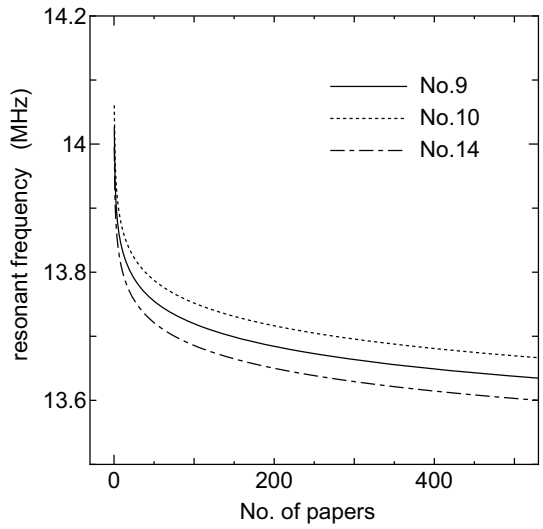

Figure 8. Best fit curves of resonant frequency of RFID tag vs number of papers

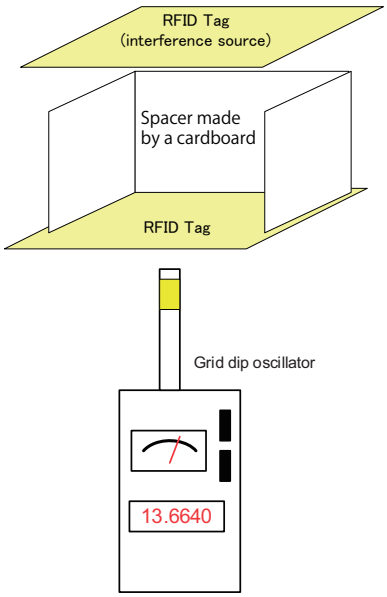

Figure 9. Measurement image of the resonant frequency of RFID tag affected by another RFID tag

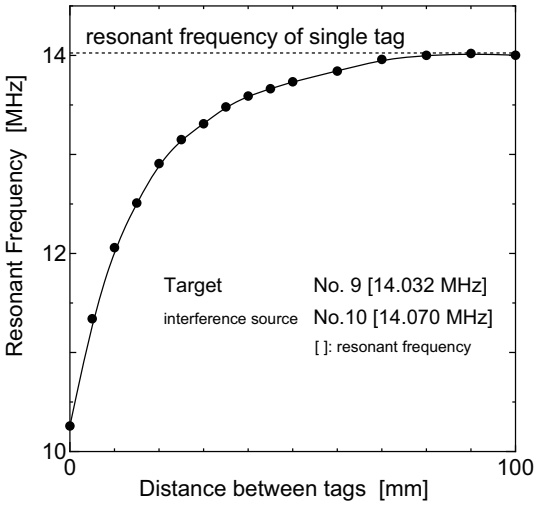

Figure 10. Resonant frequency of RFID tag vs distance between tags

evaluated. Figure 10 shows the results of this evaluation. It is clearly shown that as the distance between two pieces of tags becomes narrow, the resonant frequency decreases greatly. On the other hand, the combination between the tags is small if the distance of two tags is separated more than 7 cm.

IV. Conclusion

In this paper, using a grid dip oscillator is measured the resonant frequency of RFID tag under various conditions and is shown that as the volume of book increases, the resonant frequency becomes low. Also, the resonant frequency depends on the volume of goods and is changed like a logarithmic function. It was confirmed that the tag before attaching it on a goods in this experiment produced a resonant frequency at 14 MHz that was higher than 13.56 MHz. It is expected that it becomes an appropriate resonant frequency after attached the tag on the goods. Furthermore, it is shown that as the distance between two pieces of tags becomes narrow, the resonant frequency decreases greatly. The effect of goods to the RFID tag's performance is affected not only by the volume of goods but also environment such as the humidity. In future work, it will be investigated the influence of humidity in the performance of the RFID tag.

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